

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	An urn A contains 2 white and 4 black balls. Another urn B contains 5 white and 7 black balls. A ball is transferred from urn A to urn B. If a ball is drawn from urn B, find the probability that it is white.	5	CO1
1b	A Quantizer of a digital communication system producing the quantization error Q will have its sample values uniformly distributed and bounded by $-\Delta/2 \leq q \leq \Delta/2$ where Δ is the step size of the quantizer. Find the mean square error of the quantizer. Suggest the way to minimize the quantization error.	5	CO1
1c	Describe Gaussian distribution. Derive its mean and variance.	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	Explain joint distribution and joint density function with example.	5	CO2
2b	The joint probability density function of two random variables X and Y is given by $f_{XY}(x, y) = C(2x + y), \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 3$ $= 0, \text{ otherwise}$ Find the Value of C Also Find $P[X \geq 1, Y \leq 2]$	5	CO2
2c	Explain marginal distribution, conditional distribution and conditional expectation.	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	A random variable X has a mean of 8, variance of 9 and an unknown probability distribution function. Find $P[-4 < X < 20]$ using Chebyshev's Inequality. What conclusion you can make from this inequality theorem.	5	CO3
3b	X is a random variable with the PDF given by $f_X(x)$, and $Y = g(X)$. Derive the PDF of Y .	5	CO3
3c	The PDF of a random variable X is given by $f_X(x) = 1$, for $0 < x \leq 1$, and zero elsewhere. Determine the PDF of Y where $Y = 4X + 5$.	5	CO3
Q 4	Attempt any 2 parts of the following		
4a	Explain Wide Sense Stationary (WSS) process and its properties.	5	CO4
4b	Given a stationary random process $X(t) = 10 \cos[100t + \phi]$, where ϕ is a random variable with a uniform probability distribution in the interval $[-\pi, \pi]$. Show that $X(t)$ is ergodic in terms of Auto correlation function.	5	CO4
4c	The autocorrelation function of a random process is given by, $R_{xx}(\tau) = 36 + \frac{9}{1+6\tau^2}$. Find mean and variance of the process.	5	CO4
Q 5	Attempt any 2 parts of the following		
5a	State and prove three significant properties of Power Spectral Density	5	CO5
5b	If a WSS random process is applied as input to an LTI system, show that the output will be a WSS random process.	5	CO5
5c	Explain the significance of white Noise. Draw the plots for i) Power Spectral Density of White Noise ii) Auto correlation function of White Noise	5	CO5

Degree: B.Tech Semester: 5th CSDA - Course work**SUMMER SEMESTER (END-SEMESTER) EXAMINATION, NOV-DEC, 2025**

Course Code: CBCPC502 Course Title: Artificial Intelligence

Time: 03 Hours

Max. Marks: 50

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Explain the concept of a Rational Agent and how it differs from a purely reactive agent. Provide an example of a performance measure, environment, actuators, and sensors (PEAS) for an autonomous taxi.	5	1
1b	Define State Space Representation in the context of problem-solving in AI. How is a problem like the "Missionaries and Cannibals" typically represented using this technique?	5	1
1c	Describe the Alpha-Beta Cutoff Procedure used in game-playing algorithms. Explain the conditions under which an alpha or beta cut-off occurs and why it improves search efficiency over the standard Min-Max search.	5	1
Q 2	Attempt any 2 parts of the following		
2a	Differentiate between Informed Search and Uninformed Search strategies, providing one example algorithm for each. Explain the critical role of a heuristic function in informed search.	5	2
2b	Convert the following sentence into Predicate Logic (First-Order Logic): "All students who study hard pass the exam." Clearly identify the predicates and variables used in your conversion.	5	2
2c	Explain the concept of Conditional Independence in reasoning under uncertainty. How does this concept simplify the process of exact inference, such as through Variable Elimination, when dealing with a large number of variables?	5	2
Q 3	Attempt any 2 parts of the following		
3a	Explain the role of Resource Description Framework (RDF) in the Semantic Web. How does RDF differ from traditional relational database models in representing knowledge, and what is its fundamental structure (triples)?	5	3
3b	What is the purpose of the Web Ontology Language (OWL)? Briefly compare the representational power and complexity of OWL with simpler knowledge representation techniques like Semantic Networks or Frames.	5	3
3c	Describe the key components of a Planning System (e.g., Components of a Planning System, Goal Stack Planning). How are planning problems (Linear and Non-Linear) fundamentally different from simple search problems?	5	3

Q 4	Attempt any 2 parts of the following		
4a	Detail the steps in a typical Genetic Algorithm (GA) Cycle, starting from the initial population generation. Explain the primary function of the crossover and mutation genetic operators	5	4
4b	Differentiate between Supervised Learning and Unsupervised Learning in the context of Artificial Neural Networks (ANNs). Give one example of a task where each type of learning would be appropriate.	5	4
4c	Explain the structure and simple working principle of a Single Perceptron. What is its main limitation in terms of problems it can solve, and how does the architecture of a Multi-Layer Perceptron (MLP) address this limitation?	5	4
Q 5	Attempt any 2 parts of the following		
5a	Discuss the inherent limits of AI as a field. Provide examples of problems that are currently considered hard or potentially impossible for classical AI systems to solve completely, such as common sense reasoning or true creativity.	5	5
5b	What are the most pressing safety and ethical challenges that must be addressed to ensure AI systems are developed and deployed responsibly? Focus on the broader societal implications of advanced AI.	5	5
5c	Briefly describe two different high-level AI Architectures (e.g., purely reactive, deliberative, hybrid) and outline the primary trade-offs (e.g., speed vs. flexibility) associated with each design.	5	5